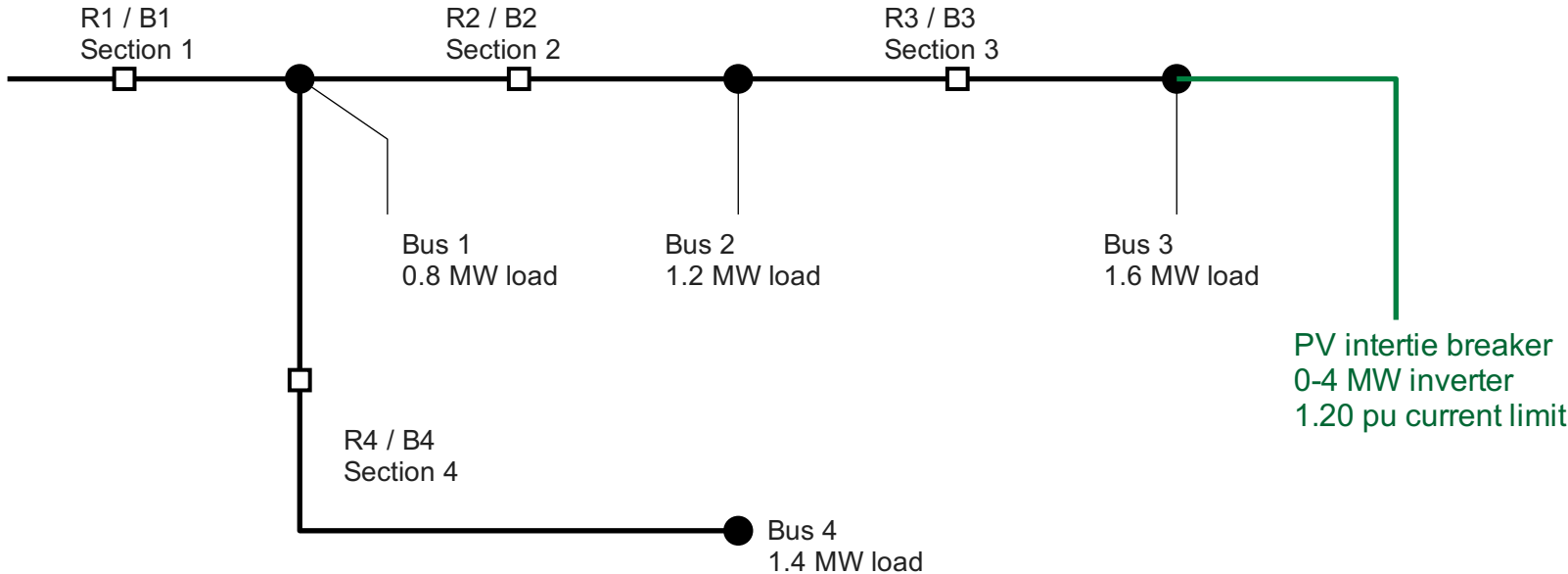


Distribution feeder single-line diagram

Utility source
11 kV, 60 Hz
Grounded Thevenin



Faults at 10%, 50%, and 90% of each section; faulted line is split at the fault node.
R1 backs up R2 and R4; R2 backs up R3. Opening B1/B2/B3 initiates direct PV transfer trip.